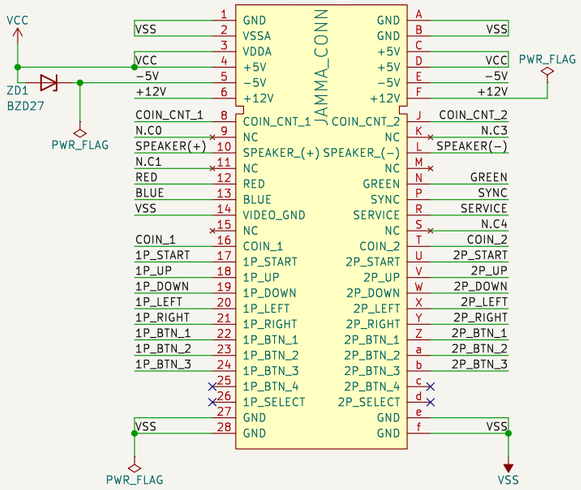
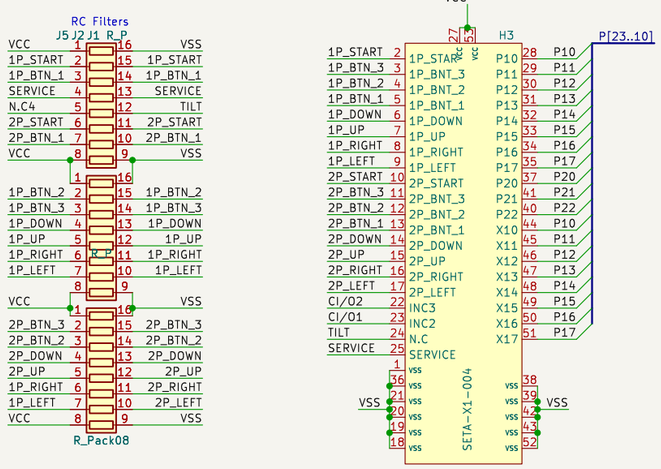
## JM1



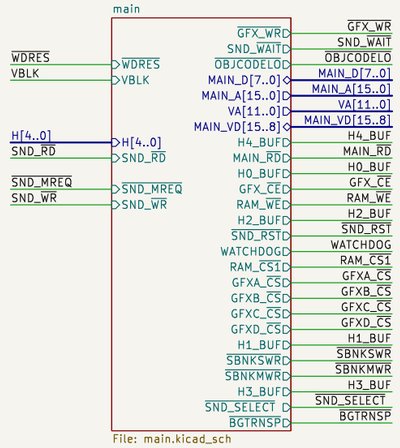

## CABINET I/O



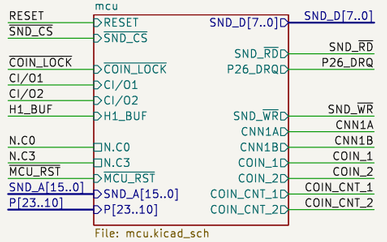




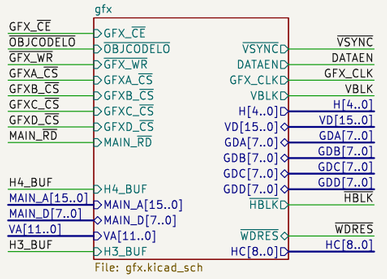
## main



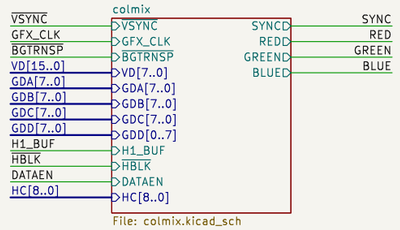

## mcu



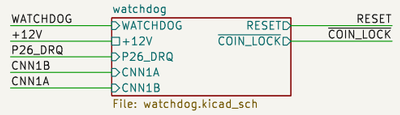

## gfx




## colmix



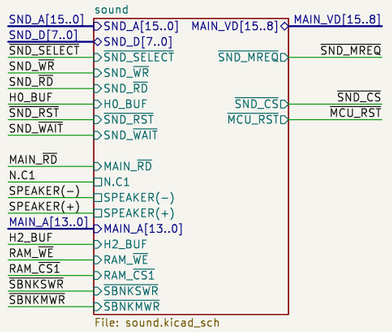

## watchdog




## capacitors




## sound

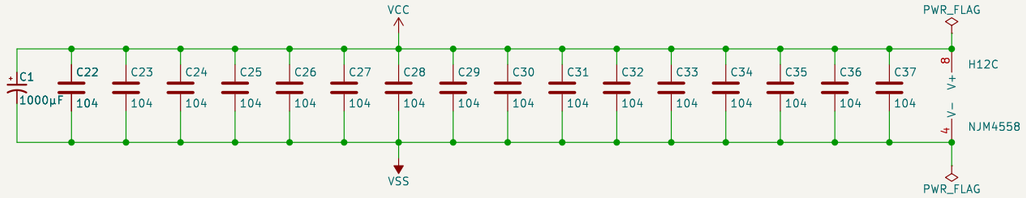



JOTEGO

Sheet: /
File: exterm.kicad_sch

**Title: EXTERMINATION**

| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 8.0.5 | | Id: 1/10 |




H5E
74LS02

B7C
74LS74

B4C
74LS74

B3C
74LS74

B2E
74LS08

G13G
74LS04

C6E
74LS32

B6E
74LS08

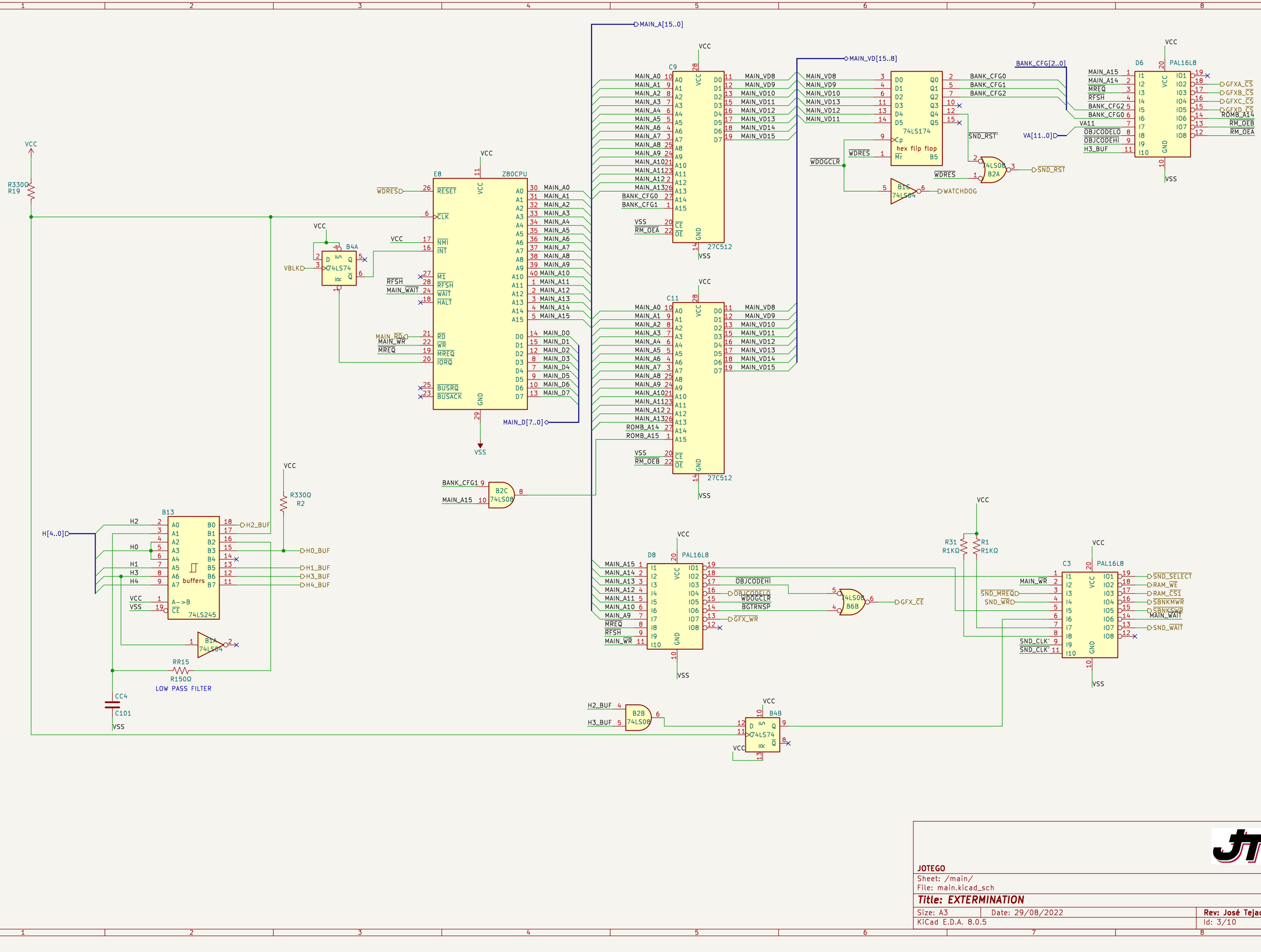
B1G
74LS04

JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

**Title: EXTERMINATION**

| | | |
|---|---|---|
| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 2/10 |

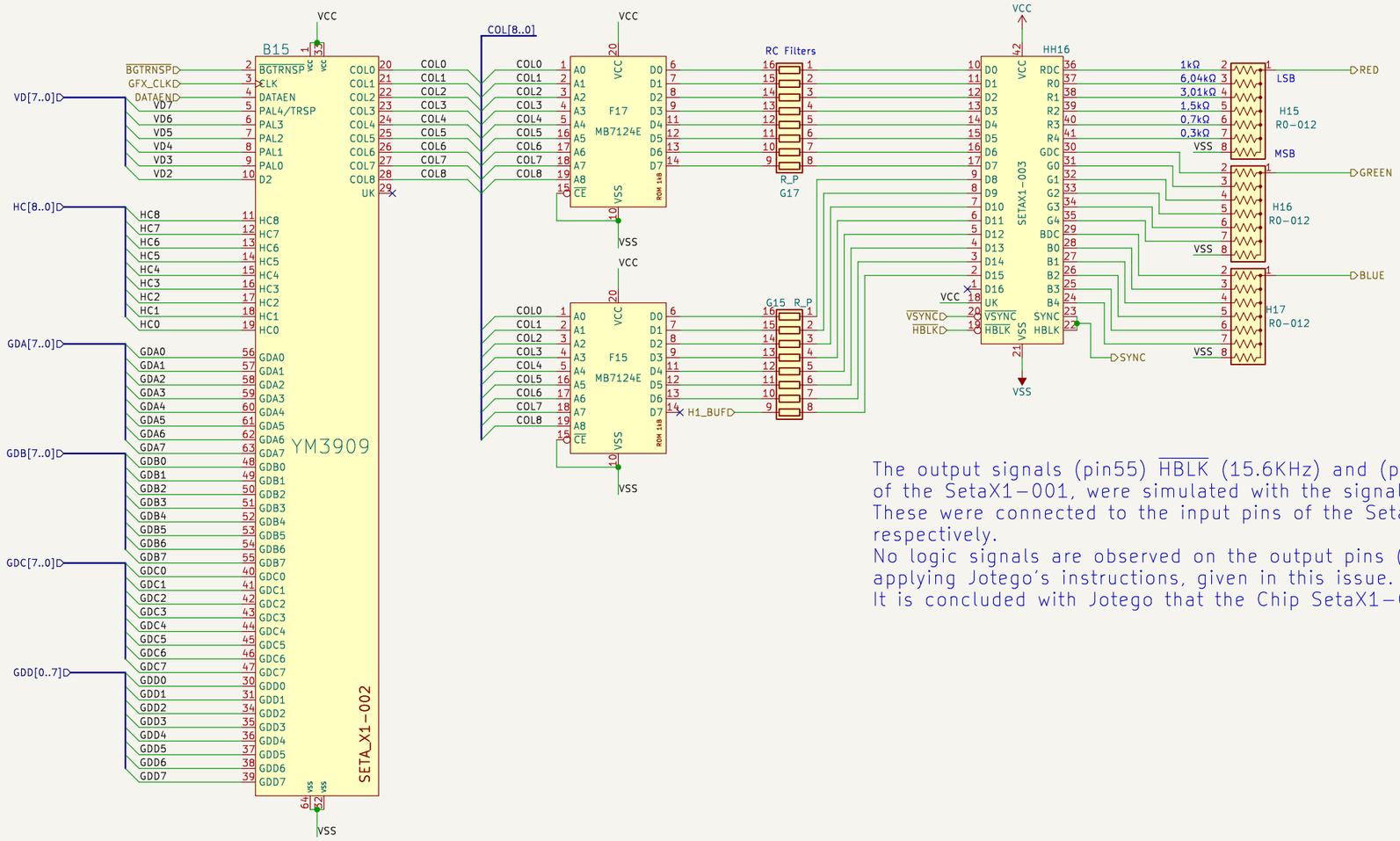



**JOTEGO**

Sheet: /main/
File: main.kicad_sch

**Title: EXTERMINATION**

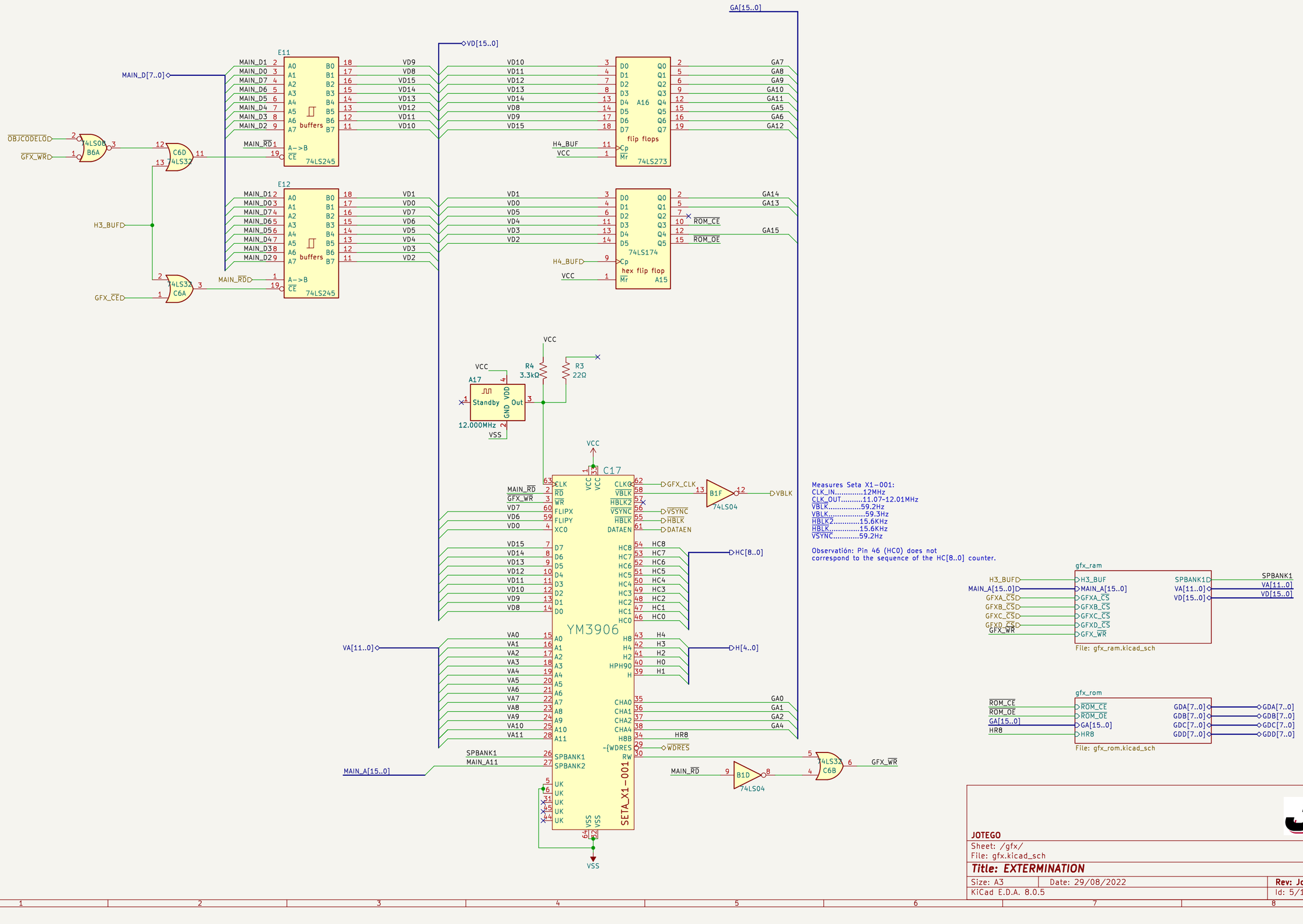
| | | |
|---|---|---|
| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 3/10 |

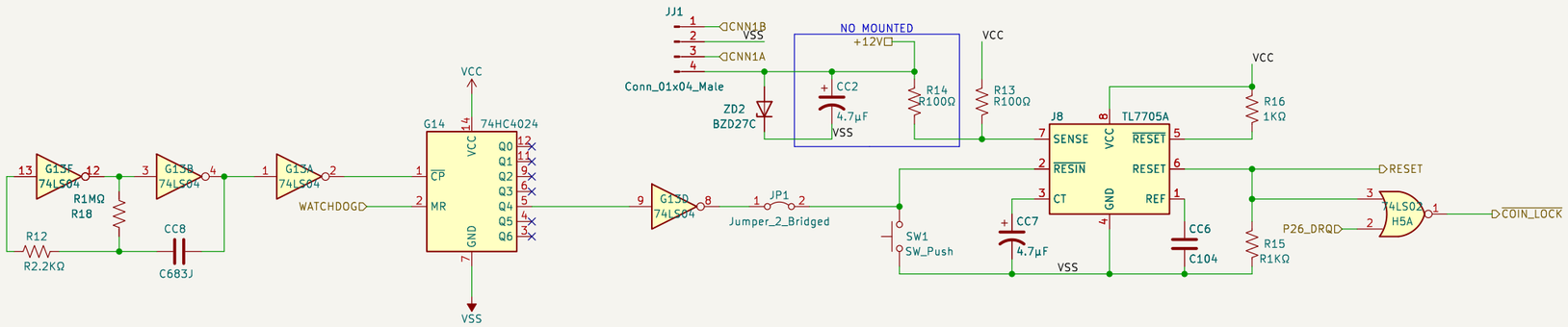



The output signals (pin55) $\overline{\text{HBLK}}$ (15.6KHz) and (pin56) $\overline{\text{VSYNC}}$ (59.2 Hz) of the SetaX1−001, were simulated with the signal generator and an inverter. These were connected to the input pins of the SetaX01−003 (pin 19 − 20), respectively.
No logic signals are observed on the output pins (22−23) HBLK − SYNC, applying Jotego's instructions, given in this issue.
It is concluded with Jotego that the Chip SetaX1−001 is broken.

JOTEGO

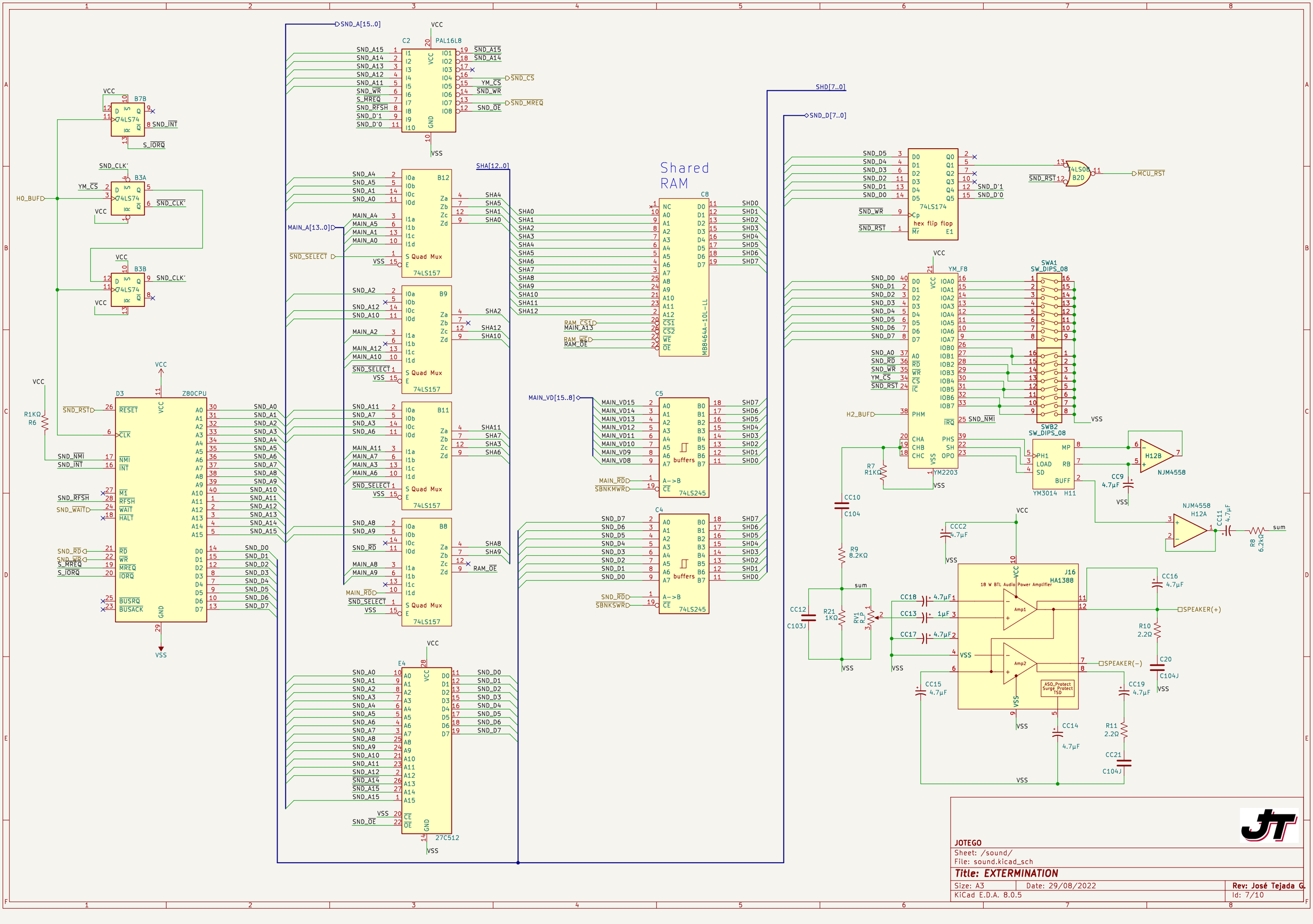
| | | |
|---|---|---|
| Sheet: /colmix/ | | |
| File: colmix.kicad_sch | | |
| **Title: EXTERMINATION** | | |
| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 4/10 |

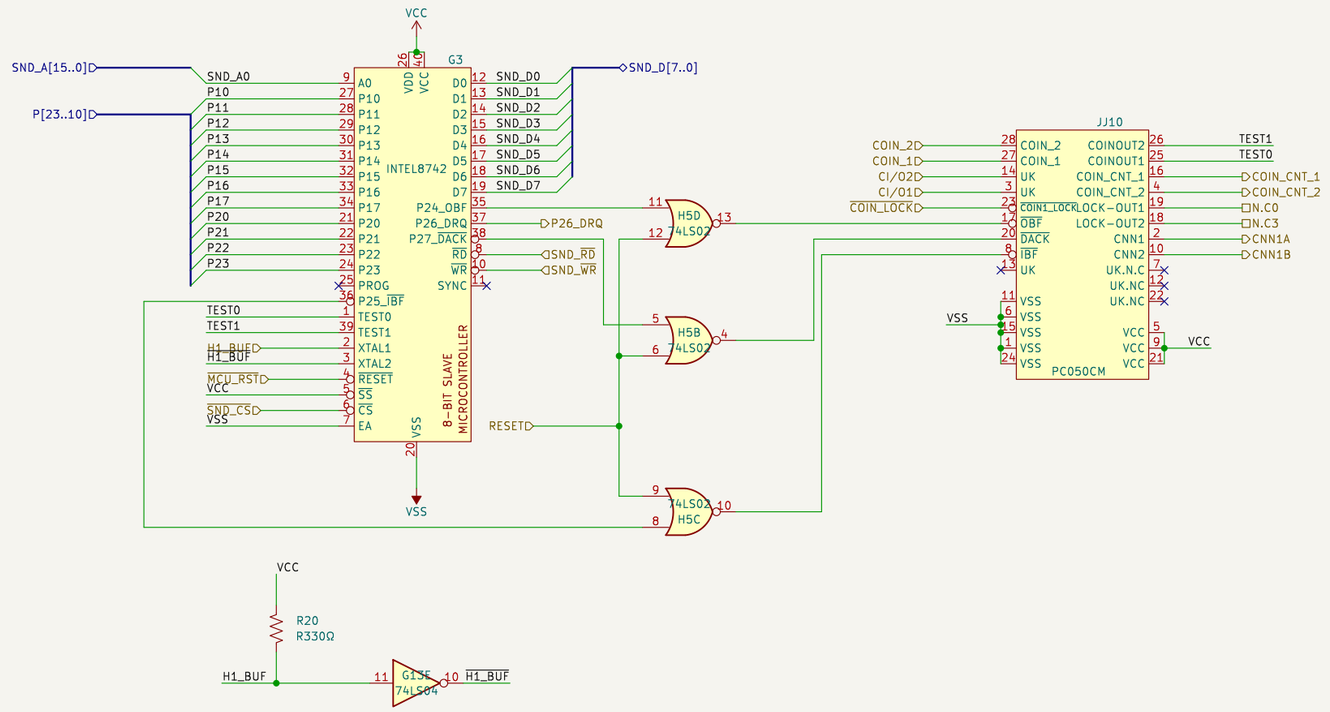

GA[15..0]
VD[15..0]
E11
E12
MAIN_D[7..0]
MAIN_D1
MAIN_D0
MAIN_D7
MAIN_D6
MAIN_D5
MAIN_D4
MAIN_D3
MAIN_D2
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
buffers
MAIN_RD
A->B
CE
74LS245
VD9 VD8 VD15 VD14 VD13 VD12 VD11 VD10
VD1 VD0 VD7 VD6 VD5 VD4 VD3 VD2
OBJCODELO
GFX_WR
74LS08
B6A
C6D
74LS32
H3_BUF
GFX_CE
C6A
D0 D1 D2 D3 D4 D5 D6 D7
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
A16
flip flops
H4_BUF
VCC
Cp
Mr
74LS273
GA7 GA8 GA9 GA10 GA11 GA5 GA6 GA12
74LS174
hex flip flop
A15
ROM_CE
ROM_OE
GA14 GA13 GA15
R4
3.3kΩ
R3
22Ω
A17
Standby
VDD
GND
Out
12.000MHz
VSS
C17
CLK
RD
WR
FLIPX
FLIPY
XC0
VCC
GFX_CLK
VBLK
HBLK2
VSYNC
HBLK
DATAEN
B1F
74LS04
HC8 HC7 HC6 HC5 HC4 HC3 HC2 HC1 HC0
HC[8..0]
YM3906
VA[11..0]
VA0 VA1 VA2 VA3 VA4 VA5 VA6 VA7 VA8 VA9 VA10 VA11
H8 H4 H2 HPH90 H
H4 H3 H2 H0 H1
H[4..0]
CHA0 CHA1 CHA2 CHA4 H8B
GA0 GA1 GA2 GA4 HR8
~{WDRES}
WDRES
RW
SPBANK1
SPBANK2
MAIN_A11
MAIN_A[15..0]
UK
SETA_X1-001
B1D
74LS32
C6B
GFX_WR
Measures Seta X1-001:
CLK_IN.....12MHz
CLK_OUT.....11.07-12.01MHz
VBLK.....59.2Hz
VBLK.....59.3Hz
HBLK2.....15.6KHz
HBLK.....15.6KHz
VSYNC.....59.2Hz
Observatión: Pin 46 (HC0) does not
correspond to the sequence of the HC[8..0] counter.
gfx_ram
H3_BUF
MAIN_A[15..0]
GFXA_CS
GFXB_CS
GFXC_CS
GFXD_CS
GFX_WR
SPBANK1
VA[11..0]
VD[15..0]
File: gfx_ram.kicad_sch
gfx_rom
ROM_CE
ROM_OE
GA[15..0]
HR8
GDA[7..0]
GDB[7..0]
GDC[7..0]
GDD[7..0]
File: gfx_rom.kicad_sch
JOTEGO
Sheet: /gfx/
File: gfx.kicad_sch
Title: EXTERMINATION
Size: A3
Date: 29/08/2022
Rev: José Tejada G.
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Id: 5/10



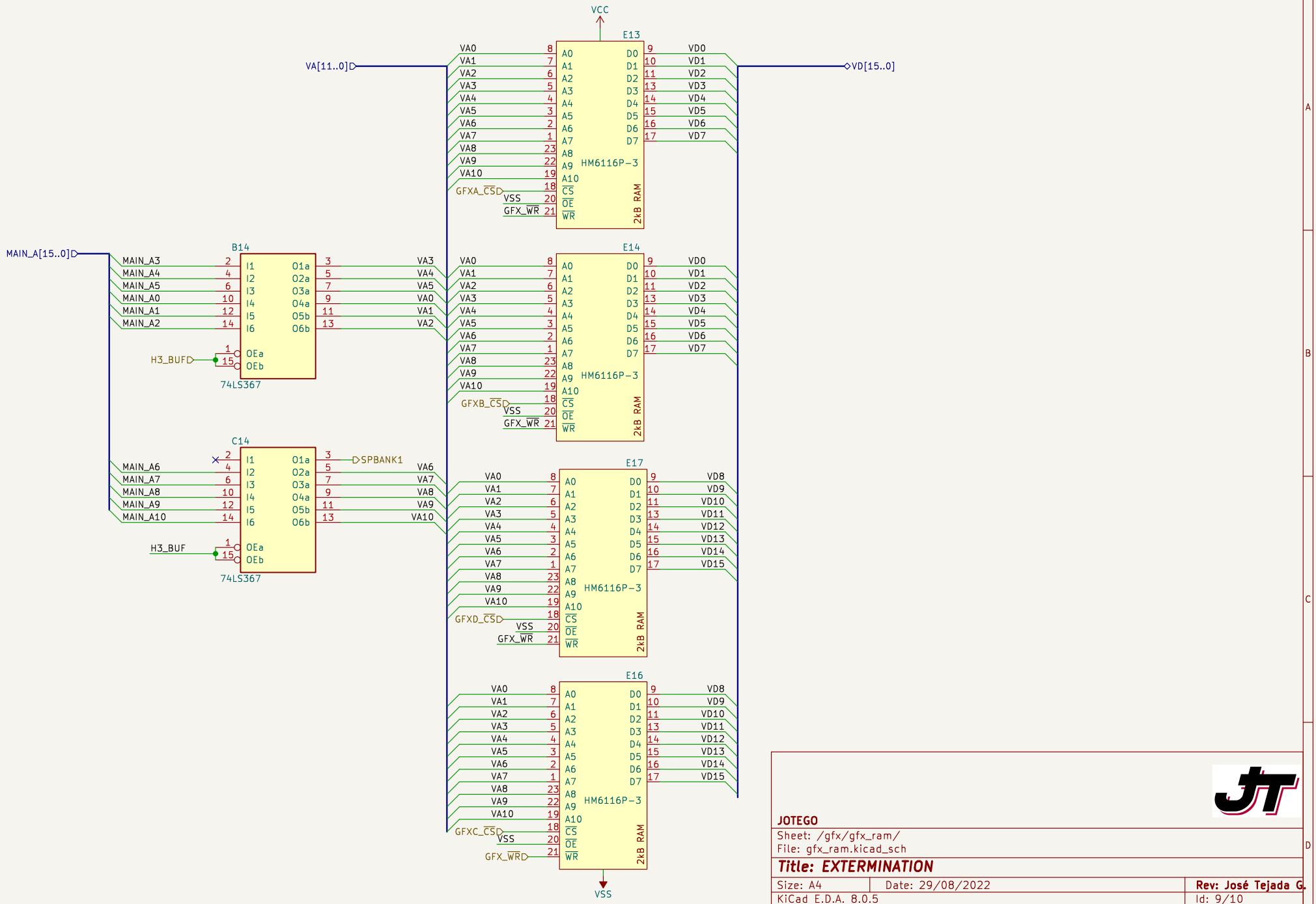

JOTEGO

| | | |
|---|---|---|
| Sheet: /watchdog/ | | |
| File: watchdog.kicad_sch | | |
| **Title: EXTERMINATION** | | |
| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 6/10 |


Shared
RAM
SND_A[15..0]
SHA[12..0]
SHD[7..0]
SND_D[7..0]
MAIN_A[13..0]
MAIN_VD[15..8]
C2 PAL16L8
B7B 74LS74
B3A 74LS74
B3B 74LS74
D3 Z80CPU
C8 MB8464A-10L-LL
C5 74LS245
C4 74LS245
B12 74LS157
B9 74LS157
B11 74LS157
B8 74LS157
E4 27C512
74LS174 hex flip flop
YM_F8 YM2203
SWA1 SW_DIPS_08
SWB2 SW_DIPS_08
H12B NJM4558
H12A NJM4558
J16 HA1388
18 W BTL Audio Power Amplifier
74LS08 B2D
ASO_Protect Surge Protect TSD
VCC
VSS
SND_CLK'
H0_BUF
H2_BUF
R1KΩ R6
R7 R1KΩ
R9 8.2KΩ
R21 1KΩ
RV1 R_Pot
R10 2.2Ω
R11 2.2Ω
R8 6.2KΩ
CC9 4.7µF
CC10 C104
CC11 4.7µF
CC12 C103J
CC13 1µF
CC14 4.7µF
CC15 4.7µF
CC16 4.7µF
CC17 4.7µF
CC18 4.7µF
CC19 4.7µF
CCC2 4.7µF
C20 C104J
C21 C104J
sum
SPEAKER(+)
SPEAKER(−)
MCU_RST
JOTEGO
Sheet: /sound/
File: sound.kicad_sch
Title: EXTERMINATION
Size: A3
Date: 29/08/2022
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 7/10




| | | |
|---|---|---|
| JOTEGO | | |
| Sheet: /mcu/ | | |
| File: mcu.kicad_sch | | |
| **Title: EXTERMINATION** | | |
| Size: A3 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 8/10 |




JOTEGO

| | | |
|---|---|---|
| Sheet: /gfx/gfx_ram/ | | |
| File: gfx_ram.kicad_sch | | |
| **Title: EXTERMINATION** | | |
| Size: A4 | Date: 29/08/2022 | **Rev: José Tejada G.** |
| KiCad E.D.A. 8.0.5 | | Id: 9/10 |

NOT POPULATED

GA[15..0]

GDA[7..0]

GDB[7..0]

GDC[7..0]

GDD[7..0]

ROM_CE 11 B1E 10 ROM_CE'
74LS04

A13 A12 A10 A8 A7 A5 A4 A2 — 27C512

| | | | |
|---|---|---|---|
| GA0 10 | A0 | D0 | 11 GDA0 |
| GA1 9 | A1 | D1 | 12 GDA1 |
| GA2 8 | A2 | D2 | 13 GDA2 |
| HR8 7 | A3 | D3 | 15 GDA3 |
| GA4 6 | A4 | D4 | 16 GDA4 |
| GA5 5 | A5 | D5 | 17 GDA5 |
| GA6 4 | A6 | D6 | 18 GDA6 |
| GA7 3 | A7 | D7 | 19 GDA7 |
| GA8 25 | A8 | | |
| GA9 24 | A9 | | |
| GA10 21 | A10 | | |
| GA11 23 | A11 | | |
| GA12 2 | A12 | | |
| GA13 26 | A13 | | |
| GA14 27 | A14 | | |
| GA15 1 | A15 | | |
| ROM_CE 20 | CE | | |
| ROM_OE 22 | OE | | |
| VCC 28 | VCC | GND | VSS 14 |

JOTEGO

Sheet: /gfx/gfx_rom/
File: gfx_rom.kicad_sch

**Title: EXTERMINATION**

| | | |
|---|---|---|
| Size: A3 | Date: 29/08/2022 | Rev: José Tejada G. |
| KiCad E.D.A. 8.0.5 | | Id: 10/10 |